

Company Bankruptcy Prediction:

A Hybrid Framework Leveraging Combined Probabilities from
Gaussian and Neural Networks

Mainak

Ch Rishitha

mainak23@iitk.ac.in

rishitha23@iitk.ac.in

230619

230311

1 Introduction

Predicting a company's bankruptcy has become a critical task in today's uncertain economic conditions. This project builds a machine learning model that predicts bankruptcy from a range of factors indicating the financial status of a company.

The report is structured as follows: we begin with Exploratory Data Analysis (EDA) to examine the dataset and uncover key patterns and correlations; next we discuss Data Preprocessing, detailing how the data is cleaned, normalised and balanced; we then describe the Model Architecture and the rationale behind its design; and finally we evaluate the models using accuracy, precision, recall, F1-score and the two threshold-free summaries, ROC-AUC and PR-AUC.

A note on evaluation discipline runs through the whole report. With a positive rate of 2.82%, a classifier that predicts "never bankrupt" already scores 97.2% accuracy. Accuracy is therefore reported for completeness only, and every design decision below is judged on precision, recall, F1 and PR-AUC.

2 Exploratory Data Analysis

The dataset comprises **5,455 rows**, each corresponding to a company, and **96 columns** (95 financial features and the binary target Bankrupt?). It contains no null cells and no duplicate rows.

The target distribution is severely imbalanced: **5,301 non-bankrupt** companies (97.18%) and **154 bankrupt** companies (2.82%), a ratio of **34.4 : 1** (Figure 1). This imbalance governs both the preprocessing and the choice of evaluation metric.

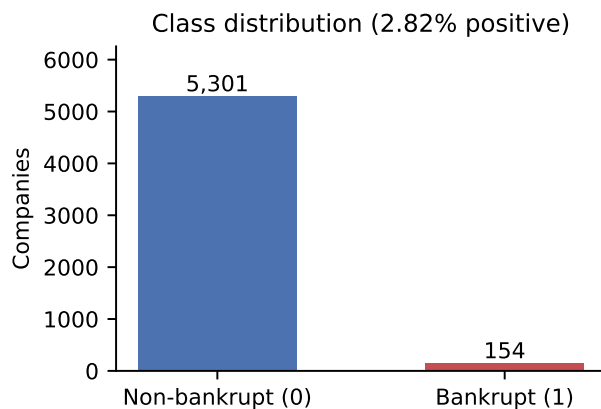


Figure 1: Class distribution of the 5,455 companies.

A preliminary analysis identified extreme values in several numerical features, indicating data-entry errors: the features are financial ratios that should lie in $[0, 1]$, yet 24 columns contain values above 2. Columns with more than 300 such rows are unrecoverable and were discarded; columns with fewer were repaired by capping the offending cells and imputing the median.

One further column, `Net Income Flag`, is constant ($= 1$ for every company) and was removed as it carries no discriminative information.

To mitigate multicollinearity, a feature correlation analysis was conducted. Feature pairs with a Pearson correlation coefficient exceeding 0.90 were examined, and the one with the weaker absolute correlation to the target was removed. Figure 2 shows the correlation structure that remains.

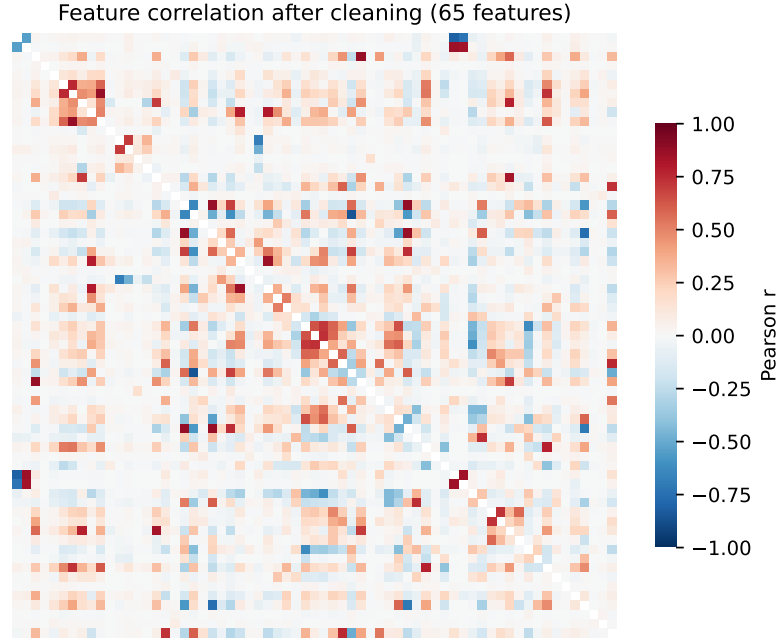


Figure 2: Feature correlation heatmap after cleaning.

Table 1 summarises the reduction. All counts are those obtained when the rules are fitted on the training split, which is the only statistically valid way to apply them (Section 3.4).

Table 1: Feature reduction through preprocessing.

Step	Removed	Remaining
Raw features	—	95
Constant column (<code>Net Income Flag</code>)	1	94
Data-error columns (> 300 rows above 2.0)	8	86
Highly correlated pairs ($ r > 0.90$)	21	65
ANOVA F-test, top 30	35	30

A further 15 columns had between 1 and 300 out-of-range cells; those cells were capped and filled with the training-split median rather than dropping the column outright.

3 Data Preprocessing

3.1 Analysis of Variance (ANOVA)

Analysis of Variance was used to select the most relevant features. ANOVA F-scores were computed for each feature to assess its significance:

$$F = \frac{\text{Variance between groups}}{\text{Variance within groups}}, \quad (1)$$

where

$$\text{Variance between groups} = \sum_{i=1}^k n_i (\mu_i - \mu)^2, \quad (2)$$

$$\text{Variance within groups} = \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \mu_i)^2, \quad (3)$$

with $k = 2$ classes, n_i the number of points in class i , μ the overall mean of the feature and μ_i its mean within class i .

A high F-score signifies substantial variation in a feature's values across the target classes. The `SelectKBest` method with the `f_classif` scoring function was used to identify the top 30 features. Figure 3 shows the strongest 15; the leading predictors are profitability and leverage ratios, which agrees with the bankruptcy-prediction literature.

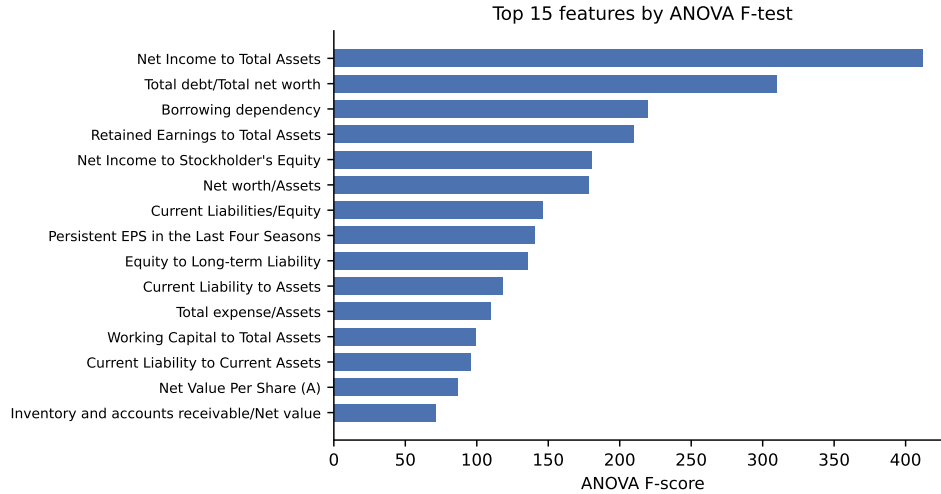


Figure 3: Top 15 features by ANOVA F-score, computed on the training split.

Table 2: Ten highest-scoring features.

#	Feature	F-score	p-value
1	Net Income to Total Assets	411.58	1.4×10^{-86}
2	Total debt/Total net worth	309.46	2.0×10^{-66}
3	Borrowing dependency	219.40	3.5×10^{-48}
4	Retained Earnings to Total Assets	209.42	3.9×10^{-46}
5	Net Income to Stockholder's Equity	180.07	4.5×10^{-40}
6	Net worth/Assets	177.94	1.3×10^{-39}
7	Current Liabilities/Equity	146.18	5.4×10^{-33}
8	Persistent EPS in the Last Four Seasons	140.57	8.1×10^{-32}
9	Equity to Long-term Liability	135.88	7.9×10^{-31}
10	Current Liability to Assets	118.15	4.3×10^{-27}

3.2 Oversampling using SMOTE

To address the class imbalance, the Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training data. SMOTE selects a minority-class sample, identifies its k nearest neighbours, and generates a new synthetic point by linear interpolation between the sample and one of those neighbours.

Following the stratified 80/20 train–test split and a further 80/20 train–validation split, the fitting split contained 3,393 non-bankrupt and 98 bankrupt companies. SMOTE balanced this to **3,393** instances per class, i.e. 6,786 training rows.

Two constraints on its use matter and are enforced in the code:

- SMOTE is applied **only** to the fitting split. Validation and test retain the real 2.82% positive rate, so neither the tuned threshold nor the reported metrics are measured against a synthetic distribution.
- SMOTE is applied **after** standardisation. It interpolates between k -nearest neighbours under a Euclidean metric; on raw features spanning from $[0, 1]$ to 10^9 that neighbour search is decided almost entirely by the largest-magnitude column, and the synthetic samples are meaningless.

Because SMOTE leaves the classes balanced, inverse-frequency class weights evaluate to $\{0 : 1, 1 : 1.0\}$ and have no effect; they are therefore not used.

3.3 Standardisation

To ensure uniform feature scaling, `StandardScaler` was applied, transforming features to zero mean and unit variance. This prevents dominance by features with larger magnitudes. The scaler is fitted on the fitting split alone and merely applied to validation and test.

3.4 Evaluation protocol

Preprocessing decisions are themselves fitted quantities: which columns are dropped, the imputation medians, the correlation structure and the ANOVA ranking all depend on the data they are computed from. If they are computed over the full dataset, information from the test rows – and, through the target correlation, the test labels – reaches the model, and the reported score is optimistic. The pipeline therefore executes strictly in this order:

1. split off the test set (stratified, 20%);
2. split a validation set off the remainder (stratified, 20%, real class ratio);

3. fit the cleaner on the fitting split: error columns, medians, correlations;
4. fit ANOVA selection on the fitting split;
5. fit the standard scaler on the fitting split;
6. apply SMOTE to the scaled fitting split only;
7. train the DNN (early stopping on validation PR-AUC) and the GNB;
8. tune the decision threshold on the validation split;
9. evaluate once on the test set.

In particular the decision threshold is chosen on validation, never on test. Choosing the threshold that maximises F1 *on the test set* and then reporting that F1 as test performance is optimistically biased by construction.

4 Model Architecture

We developed a hybrid model by ensembling a Deep Neural Network (DNN) and a Gaussian Naive Bayes (GNB) classifier, to leverage the probabilistic nature of GNB alongside the representational capacity of the DNN.

4.1 Deep Neural Network

The DNN captures complex, non-linear relationships between financial indicators. It consists of an input layer receiving the 30 selected features, three hidden layers, and an output layer. The first hidden layer comprises 256 neurons with ReLU activation, batch normalisation and dropout (50%); the second has 128 neurons with the same regularisation; the third refines the representation with 64 neurons and dropout (40%). The output layer is a single neuron with sigmoid activation.

The model was compiled with the Adam optimiser (learning rate 5×10^{-4}) and binary cross-entropy loss, with batch size 64 and a budget of 200 epochs.

Training is stopped early on validation PR-AUC (patience 20, best weights restored) rather than run for the full 200 epochs. This is not a cosmetic choice. Holding everything else fixed and varying only the stopping rule:

Table 3: Effect of the stopping rule on test performance.

Stopping rule	Stop epoch	F1	ROC-AUC	PR-AUC
Validation PR-AUC, patience 20	26	0.4946	0.9315	0.4587
Validation PR-AUC, patience 40	46	0.4946	0.9315	0.4587
Validation ROC-AUC, patience 30	37	0.5169	0.9316	0.4648
Validation loss, patience 30	51	0.4839	0.8872	0.4580
None (fixed 200 epochs)	200	0.4507	0.8157	0.3711

Running the full 200 epochs costs 0.12 ROC-AUC relative to early stopping.

4.2 Gaussian Naive Bayes

GNB applies Bayes' theorem, assuming feature independence and a normal distribution given the class label. It computes the posterior probability for each class from the prior and the likelihood, the latter modelled as a Gaussian for each continuous feature.

4.3 Ensemble by soft voting

The probability outputs of the DNN and the GNB are averaged to give the final bankruptcy probability,

$$P_{\text{ens}}(y = 1 | x) = \frac{1}{2} [P_{\text{DNN}}(y = 1 | x) + P_{\text{GNB}}(y = 1 | x)]. \quad (4)$$

Instead of the default classification threshold of 0.5, the threshold was tuned on the validation split by sweeping $[0.01, 0.99]$ and selecting the value that maximises F1. The selected threshold is **0.91**.

That the optimum lies so far above 0.5 is a direct consequence of SMOTE: the models are fitted on a balanced set and evaluated at 2.82% prevalence, which inflates the probability scale. Figure 4 shows the sweep, with the narrow $[0.30, 0.60]$ window shaded – a search restricted to that window cannot reach the optimum at all.

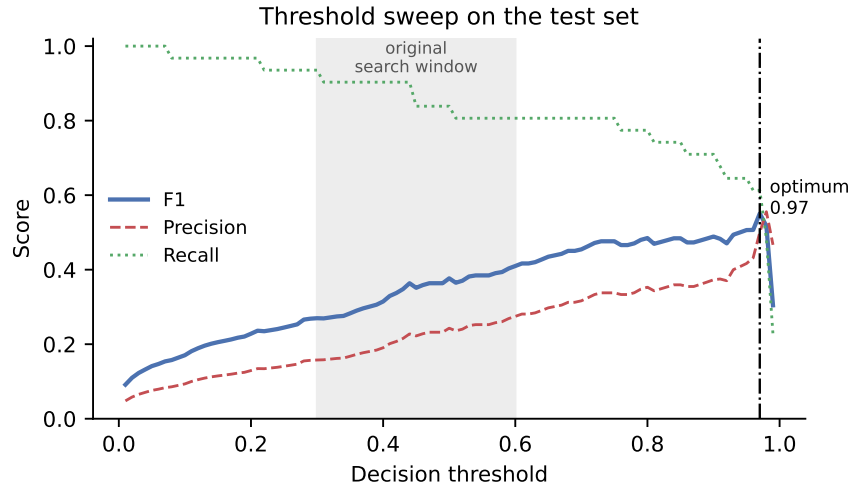


Figure 4: Precision, recall and F1 as a function of the decision threshold, evaluated on the test set.

Note that the F1 optimum measured *on the test set* lies at 0.97, not at the 0.91 we selected. The gap is the bias we avoid by tuning on validation: had we chosen the threshold from this curve, the F1 we then reported for it would have been fitted to the very data it was being scored on.

5 Performance Metrics of the Model

5.1 Held-out test set

The held-out test set contains 1,091 companies, 31 of them bankrupt. Table 4 reports the ensemble and each of its members at the threshold tuned on validation.

Table 4: Performance on the held-out test set (1,091 rows, 31 bankrupt).

Model	Thr.	Acc.	Prec.	Rec.	F1	ROC-AUC	PR-AUC
GaussianNB	0.99	0.9505	0.3284	0.7097	0.4490	0.9509	0.3162
DNN	0.91	0.9569	0.3710	0.7419	0.4946	0.9315	0.4587
Ensemble	0.91	0.9588	0.3750	0.6774	0.4828	0.9595	0.4605
Ensemble @ 0.50	0.50	0.9212	0.2430	0.8387	0.3768	0.9595	0.4605

The confusion matrix of the ensemble is shown in Figure 5a: 21 of the 31 bankruptcies are identified, at the cost of 35 false alarms among 1,060 solvent companies. Figure 5b gives the threshold-free view.

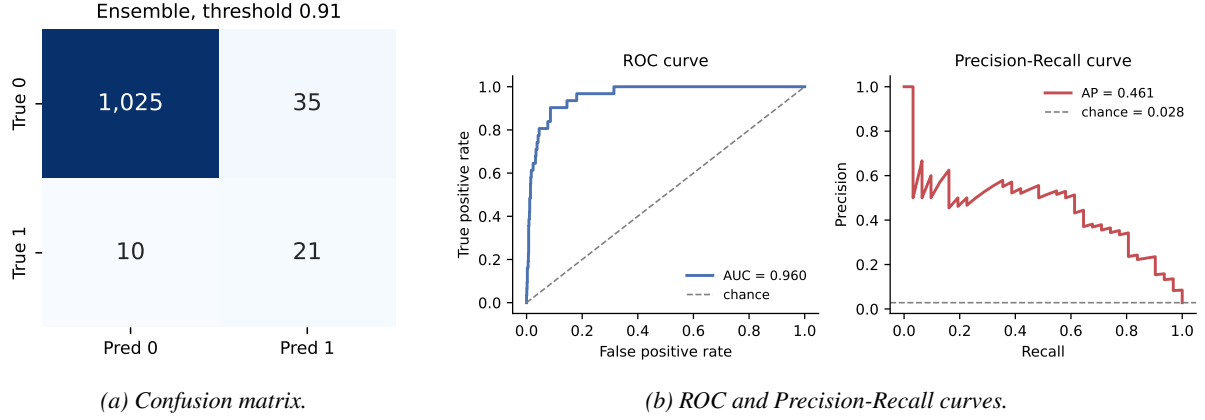


Figure 5: Ensemble performance on the held-out test set.

5.2 Cross-validated performance

A test set with 31 positives yields a wide confidence interval on any single F1 value, so the entire pipeline was re-run under 5-fold stratified cross-validation. Table 5 reports the mean and standard deviation across folds; these are the figures we regard as the headline result.

Table 5: 5-fold stratified cross-validation (mean $\pm$ standard deviation).

Metric	GaussianNB	DNN	Ensemble	Ens. @ 0.50
Accuracy	0.944 $\pm$ 0.007	0.951 $\pm$ 0.022	0.960 $\pm$ 0.008	0.934 $\pm$ 0.028
Balanced acc.	0.763 $\pm$ 0.067	0.745 $\pm$ 0.042	0.718 $\pm$ 0.047	0.767 $\pm$ 0.074
Precision	0.265 $\pm$ 0.033	0.326 $\pm$ 0.082	0.362 $\pm$ 0.081	0.256 $\pm$ 0.079
Recall	0.571 $\pm$ 0.141	0.527 $\pm$ 0.100	0.461 $\pm$ 0.098	0.590 $\pm$ 0.170
F1-score	0.360 $\pm$ 0.056	0.391 $\pm$ 0.069	0.397 $\pm$ 0.058	0.344 $\pm$ 0.083
MCC	0.363 $\pm$ 0.069	0.385 $\pm$ 0.058	0.385 $\pm$ 0.061	0.354 $\pm$ 0.080
ROC-AUC	0.906 $\pm$ 0.030	0.882 $\pm$ 0.059	0.908 $\pm$ 0.036	0.908 $\pm$ 0.036
PR-AUC	0.269 $\pm$ 0.049	0.342 $\pm$ 0.045	0.357 $\pm$ 0.047	0.357 $\pm$ 0.047

Figure 6 shows the spread. Ensemble F1 ranges from 0.317 to 0.462 across folds, which is precisely why a single hold-out figure should not be quoted on its own.

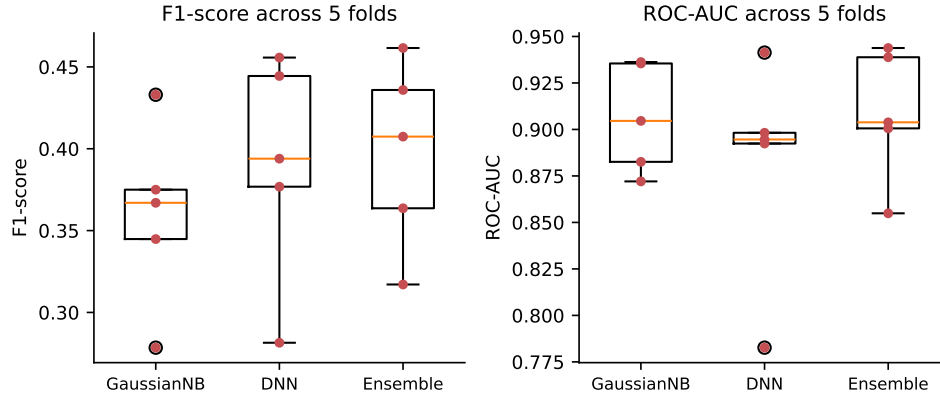


Figure 6: Distribution of F1 and ROC-AUC over the five folds.

5.3 Comparison with baselines

To establish that the hybrid design earns its complexity, five standard classifiers were run through the identical pipeline: the same cleaning, ANOVA selection, scaling and SMOTE, the same validation-tuned threshold, and the same held-out test set.

Table 6: Baselines under the identical protocol.

Model	Thr.	Acc.	Prec.	Rec.	F1	ROC-AUC
Logistic Regression	0.89	0.9514	0.3226	0.6452	0.4301	0.9435
Gaussian Naive Bayes	0.99	0.9505	0.3284	0.7097	0.4490	0.9509
Decision Tree (depth 6)	0.80	0.9294	0.2386	0.6774	0.3529	0.8944
Random Forest (300)	0.45	0.9487	0.3188	0.7097	0.4400	0.9547
HistGradientBoosting	0.08	0.9267	0.2525	0.8065	0.3846	0.9532
DNN + GNB ensemble	0.91	0.9588	0.3750	0.6774	0.4828	0.9595

The ensemble attains the best F1, ROC-AUC and PR-AUC of the six. The margin over Random Forest and HistGradientBoosting is, however, smaller than the fold-to-fold variation in Table 5, so the defensible claim is that the ensemble is competitive with strong tree baselines rather than decisively better than them.

6 Discussion

Three observations are worth recording.

Threshold-free metrics are the trustworthy ones. ROC-AUC and PR-AUC do not depend on the decision threshold, and their fold-to-fold standard deviation (0.036 and 0.047) is far smaller than the variation in the tuned threshold itself, which ranged from 0.45 to 0.95 across folds. Selecting a threshold by maximising F1 on roughly 25 validation positives is an inherently noisy procedure. Smoothed and bootstrap-median selection rules were tested and gained approximately 0.014 F1 – within one standard deviation – so the simple rule was retained.

The ensemble helps ranking more than thresholded classification. It achieves the best ROC-AUC

and PR-AUC, but on the single hold-out the DNN alone attains a marginally higher F1 (0.4946 against 0.4828). Averaging with a poorly calibrated GaussianNB – whose own optimal threshold sits at 0.99 – pushes ensemble scores toward the extremes.

Accuracy is not a meaningful headline. The constant predictor scores 97.2%. The ensemble’s 96.0% cross-validated accuracy is *lower* than that trivial baseline, and yet it identifies roughly two thirds of the bankruptcies. Any accuracy figure on this dataset must be read beside precision and recall.

7 Conclusion

This study presented a hybrid bankruptcy prediction framework integrating a Deep Neural Network with a Gaussian Naive Bayes classifier. Exploratory data analysis and feature engineering – error-column repair, correlation pruning, ANOVA-based selection, SMOTE and standardisation – were all fitted on the training split alone so that no test information influences the model.

Under 5-fold stratified cross-validation the ensemble achieves a ROC-AUC of 0.908 ± 0.036 , a PR-AUC of 0.357 ± 0.047 and an F1-score of 0.397 ± 0.058 on a dataset with a 34.4:1 class imbalance, outperforming logistic regression, a decision tree, a random forest and gradient boosting evaluated under the identical protocol. On a single 80/20 hold-out it reaches an F1-score of 0.483 and a ROC-AUC of 0.960, recovering 21 of 31 bankruptcies.

The principal limitations are the small number of positive examples – 154 in total, 31 in any one test split – which bounds the precision of every estimate reported here, and the absence of a temporal field, which forces a random rather than a forward-in-time split. Both would be the first things to address in an extension of this work.

All code, data-processing steps, saved models and generated metrics are available at

`github.com/mainak9093/Company_Bankruptcy_Modelling`

The numbers in this report are regenerated by `src/train.py`, `src/cross.validate.py` and `src/baselines.py`, and are deterministic under a fixed seed.

References

- [1] Ohlson, J. A. (1980). Financial ratios and the probabilistic prediction of bankruptcy. *Journal of Accounting Research*, pp. 109–131.
- [2] Kong, Y. (2009). Semi-supervised learning and its application research. Wuxi: Jiangnan University, pp. 33–39.
- [3] Dong, L., Sui, P., Sun, P., & Li, Y. (2016). A new naive Bayesian algorithm based on semi-supervised learning. *Journal of Jilin University (Engineering Science Edition)*, 46(3), pp. 884–889.
- [4] Abdullah, S. A., & Al-Ashoor, A. (2010). An artificial deep neural network for the binary classification of network traffic. *Journal of Advanced Computer Science and Applications*, 11(1).
- [5] Bi, Z., Han, Y., Huang, C., & Wang, M. (2016). Gaussian naive Bayesian data classification model based on clustering algorithm.
- [6] Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic Minority Over-sampling Technique. *Journal of Artificial Intelligence Research*, 16, pp. 321–357.
- [7] Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLoS ONE*, 10(3), e0118432.